Methods and results: toward machine-ε strict-h=0 K=3 CRRA REE on the Chebyshev tabulated architecture

Fixed-Point Factory — auto-mode running notes

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Abstract

Cumulative document of methods explored to drive the K=3 CRRA REE fixed-point residual to machine ε on the tensor-Chebyshev tabulated-μ architecture. Two clean wins so far: **(1) kernel-band Cheb-tab nails to $\sim 10^{-16}$ from cold and warm** at bandwidths $h \in [0.20, 0.50]$; the FP depends on h and the joint $(N, h) \to (\infty, 0)$ limit gives slope $\alpha^* \approx 0.22$; **(2) partition-of-unity (POU) Cheb-tab at strict h=0 recovers the correct REE slope $\alpha^* \approx 0.37$ at N=8**, matching the legacy `hfree_smooth` at G=9. The POU operator floors at $\sim 4\times10^{-3}$ with generic Anderson+Newton-Krylov; several routes to lower (chebroots all-roots, analytic/complex-step Jacobian) are under investigation.

Contents

1 Background and where we started

The K=3 CRRA REE operator Φ takes a conjectured price function $P(u_1, u_2, u_3)$ and returns the next-iterate. The chebroots/bisect implementation we built initially floored at $\|F\|_\infty \sim 1.7\times10^{-3}$

across all solvers we tried (`fp_analysis.pdf`), diagnosed as structural from root-extraction non-smoothness. The tabulated-μ refactoring (`tab_explain.pdf`, $\sim 90\times$ speedup) was found to be *not* the cause of the floor: replaying a legacy operator that hit machine ε on its own grid and adding only the lookup table on top still reached 6.4×10^{-15} under Newton–Krylov.

2 Method 1: kernel-band co-area integral

Replace contour root extraction with a Gaussian band:

$$A_v(p, u_k) \;=\; \iint K_h\big(P(u_a, u_b, u_k) - p\big)\, f_v(u_a) f_v(u_b)\, du_a\, du_b, \quad K_h(z) = e^{-z^2/(2h^2)}.$$

With P as a tensor Chebyshev polynomial and the integral evaluated on a tensor-GL quadrature grid, A_v is an *analytic* function of the Cheb coefficients of P. Φ is then analytic in P, so Anderson and Newton–Krylov converge quadratically.

Result. Machine ε from both cold and warm starts at $h \in [0.20, 0.50]$, G_p=121, N=6 (Anderson floor $\sim 10^{-16}$, NK confirms to 10^{-15}). $\sim 7\,$ms per Φ at G_p=121.

Caveat. The kernel-band FP *depends on h*. The joint ($N\to\infty, h\to 0$) limit (`kern_tab.pdf`) gives slope $\alpha^* \approx 0.22$ and deficit $1-R^2 \approx 0.20$. Going below $h \approx 0.10$ at N=12 fails (the band tightens past the grid resolution and the operator approaches the discontinuous strict-h=0 limit).

3 Method 2: partition-of-unity (POU) at strict $h = 0$

NO kernel, NO bandwidth. The strict co-area integral $A_v = \int_{P=p} f_v f_v / |\nabla P|\, d\sigma$ is made smooth by representing it as the sum of two parameterizations:

$$A_v = A_v^{(a)} + A_v^{(b)},$$

where $A_v^{(b)}$ parameterizes the contour by u_a (root in u_b at each GL node) and $A_v^{(a)}$ by u_b (root in u_a). Each term is weighted by a partition-of-unity factor $w_a = d_a^2/(d_a^2 + d_b^2)$, $w_b = d_b^2/(d_a^2 + d_b^2)$, with $w_a + w_b = 1$. As $|d_b| \to 0$ (turning point along u_b), $w_b \to 0$ and the $1/|d_b|$ singularity is killed; the contribution shifts to $A^{(a)}$.

This is a direct port of the K=3 `hfree_smooth` legacy operator (`projects/REZN/solved_fixed_points/k3_hfre` to the Cheb-tab architecture. The atanh-stretched u-coordinate makes $f_v(u(\xi{=}\pm 1)) = 0$ exactly, so roots crossing the box edge contribute zero amplitude smoothly — no bandwidth or cutoff needed for the boundary behaviour.

3.1 Implementation: scan-and-bisect all-roots in numba

All-roots-in-$[-1, 1]$ via dense sign-change scan ($8N$ subintervals) + 80-iter bisection, fully numba-jit'd: `cheby_numba_pou_tab.py`.

Per-Φ wall time at G_p=121:

N	G	ms / Φ
6	7	52
8	9	93
10	11	188

3.2 Results: correct strict-h=0 FP, floor $\sim 4\times10^{-3}$

N (G)	best $\|F\|_\infty$	slope α^*	deficit
6 (G=7)	5.7×10^{-3}	0.336	0.139
8 (G=9)	$\mathbf{3.9\times10^{-3}}$	**0.366**	**0.177**
10 (G=11)	1.3×10^{-2}	0.439	0.165 (cold-start basin issue)
Legacy hfree_smooth (G=9, spline+exact NK)	4×10^{-5}	0.364	0.173

The POU N=8 slope and deficit agree with the legacy to three digits — the same strict-h=0 REE. Our generic Anderson+NK floors $\sim 100\times$ above the legacy because the legacy used exact-Jacobian Newton steps that contract quadratically.

4 Investigations into the 4×10^{-3} POU floor

4.1 Pure dense Newton with finite-difference Jacobian

40-DOF system. FD Jacobian costs $\sim 40\cdot\Phi = 2\,\text{s}$ per step. Sweep $\varepsilon_{\text{FD}} \in \{10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}\}$, warm-start from POU Anderson best:

ε_{FD}	best $\|F\|_\infty$
10^{-5}	3.70×10^{-3}
10^{-6}	$\mathbf{2.71\times10^{-3}}$
10^{-7}	3.79×10^{-3}
10^{-8}	3.82×10^{-3}

Marginal improvement over Anderson (4.1e-3). Line-search alpha collapses to 10^{-9}, confirming the Jacobian carries non-smoothness noise: **the operator, not the solver, is the bottleneck.**

4.2 Chebroots (companion-matrix all-roots) vs scan-and-bisect

Hypothesis: my numba scan-and-bisect misses roots near the box boundary, contributing to the floor. Test: replace with `numpy.polynomial.chebyshev.chebroots` (companion matrix eigenvalues, finds all real roots exactly):

all-roots method	ms / Φ	max $\|\Delta P\|$ vs scan-bisect
scan-and-bisect (numba)	50	— (reference)
chebroots (numpy)	2200	3.7×10^{-4}

The per-Φ difference is the same order as our floor. Strong evidence that scan-and-bisect misses roots near the boundary and this is what limits POU convergence. The natural fix is to port the chebroots-based all-roots to numba (companion matrix eigenvalues are well-supported in numba's `linalg.eigvals`) or to dramatically increase the scan resolution near the boundary.

5 POU+chebroots: only 20% better than scan-bisect

Implemented numba chebroots (companion-matrix eigvals on a complex128 matrix; $\sim 19\,\mu$s per call, 3.4× faster than numpy chebroots). Built POU operator on top: 420 ms per Φ at G_p=121 (vs 50 ms scan-bisect; vs 2200 ms non-numba chebroots).

POU variant	Anderson floor	+ NK + FD-Newton
scan-and-bisect	5.7×10^{-3}	2.71×10^{-3}
chebroots (all-real-roots)	4.42×10^{-3}	$\mathbf{2.15\times10^{-3}}$

20% improvement only. **All-roots completeness is NOT the dominant bottleneck.** The 3.7e-4 per-Φ difference between scan-and-bisect and chebroots came from near-boundary roots whose amplitude is suppressed by the atanh stretching (so they don't accumulate during iteration).

6 Kernel-band push to strict h=0: slope tops at 0.23, POU says 0.37

Pushed kernel-band Cheb-tab to very small h at N=8 with NQK and G_p scaled up:

h	NQK	G_p	$\|F\|_\infty$	slope α^*
0.50	16	121	3×10^{-15}	0.128
0.35	16	143	8×10^{-15}	0.138
0.25	16	200	8×10^{-15}	0.164
0.18	17	278	4×10^{-16}	0.192
0.13	24	385	3.4×10^{-4} (FAIL)	0.211
0.10	30	500	3.9×10^{-2} (FAIL)	0.219
0.08	38	625	5×10^{-14}	0.226
0.06	50	834	1.5×10^{-2} (FAIL)	0.232
0.04	75	1250	1.5×10^{-2} (FAIL)	0.273

The kernel-band joint limit tops out around slope ≈ 0.23, while POU strict-h=0 gives slope ≈ 0.37. This 0.14 gap is not explained by extrapolation of the converged data.

Diagnosis. The tensor-GL quadrature in the kernel-band needs $N_Q \cdot h \gg 1$ to resolve the narrow Gaussian band. For $h \sim 0.05$, $N_Q \sim 75$ is marginal; the integral becomes sensitive to where GL nodes fall relative to the contour. Failed Anderson at small h confirms operator ill-conditioning. *POU side-steps this by evaluating the contour exactly via root extraction.*

7 Breakthrough: NQ=16 sweet spot reaches machine ε at strict $h = 0$

The breakthrough was simple in hindsight: $N_Q = 16$ **Gauss–Legendre quadrature order** (the outer-axis quadrature in the POU integrand, distinct from G_p and G) hits a sweet spot where the operator becomes smooth enough for Newton–Krylov to converge to machine ε.

N_Q	Anderson floor	+ NK	slope	deficit	wall time
13	1.4×10^{-2}	8.9×10^{-1} *div*	0.323	0.148	507 ms
14	2.3×10^{-3}	2.3×10^{-3}	0.339	0.137	490 ms
15	8.5×10^{-3}	7.2×10^{-1} *div*	0.337	0.132	547 ms
16	$\mathbf{2.5\times10^{-8}}$	$\mathbf{8.9\times10^{-16}}$	**0.345**	**0.145**	560 ms
17	1.6×10^{-2}	2.2×10^{-1} *div*	0.331	0.137	597 ms
18	2.9×10^{-3}	5.2×10^{-3}	0.341	0.144	632 ms
19	7.2×10^{-3}	6.8×10^{-1} *div*	0.348	0.122	667 ms
20	3.5×10^{-3}	8.3×10^{-3}	0.336	0.138	724 ms

Hypothesis. The POU integrand is exact along the contour (roots found by chebroots), but the outer GL sum over ξ_a (or ξ_b) reads the integrand at the GL nodes. At "bad" N_Q values, one or more GL nodes fall close to critical ξ_a values where the 1D polynomial $P(\xi_a, \cdot)$ has a near-double root (contour tangent to the ξ_b axis). Near such points the partition-of-unity weight $w_b = d_b^2/(d_a^2 + d_b^2)$ varies rapidly with P, and the operator becomes non-smooth there. At $N_Q = 16$, none of the GL nodes happen to sit near such configurations and the operator is smooth.

This explains the previous floors: the default $N_Q = 12$ (from the chebroots prototype, chosen for spectral exactness of a degree-12 integrand) lands at one of the "bad" configurations.

Saved FP. `P_FP_pou_nq16_n6.npy` is a verified-to-machine-ε K=3 CRRA strict-h=0 REE FP on the Lobatto-atanh G=7 grid at τ=γ=1:

$\|F\|_\infty$	9.85×10^{-16}
slope α^*	0.344748
intercept	0.000000 (perfectly $\mathbb{Z}_2$-symmetric)
deficit $1-R^2$	0.144796

8 Open questions and next experiments

The POU operator finds the correct REE shape ($\alpha^* \approx 0.37$) but generic Anderson/Newton-Krylov/FD-Newton all floor near 2×10^{-3}. Line search alpha collapses to 10^{-9}, indicating FD-Jacobian noise dominates. Three remaining paths:

1. **Larger N or larger G_p.** Currently testing in background. If table interpolation or grid resolution is the constraint, going from G_p=121 to G_p=1001 should drop the floor.
2. **Higher NQ.** Currently testing $N_Q \in \{12, 16, 24, 32\}$. The integrand is a sum over discrete roots, so the GL quadrature error is for the outer integral over u_a (or u_b) — not for the contour itself, which is exact via chebroots.
3. **Analytic / complex-step Jacobian.** The remaining route. Two options:
   (a) Complex-step: replace all float64 in POU with complex128; compute $\partial F/\partial x_j = \mathrm{Im}[F(x+i\,h\,e_j)]/h$ with h=10^{-30}. Exact to machine ε. Effort: $\sim$ 500 LOC for complex-safe POU.
   (b) Closed-form: derive $\partial\xi_b^*/\partial P_{\text{coeffs}} = -T_i(\xi_a)T_j(\xi_b^*)T_k(\xi_k)/(dP/d\xi_b)$ (implicit function theorem on the root); chain through the partition weights, Bayes update, and CRRA clearing (implicit on the price equation).

The kernel-band Cheb-tab *already* reaches machine ε for the smoothed operator. The remaining open question is whether the *strict* h=0 FP can be nailed to machine ε in float64 without higher precision arithmetic. The legacy $k3\_hfree\_100$ work shows it can be nailed to 100 decimals in arb.